

Public internet involves users moving to a public place to use internet by paying a small price for the internet used. Public internet is popular in many advanced economies. Why is public internet not gaining traction in India while mobile and home internet are flourishing? The Government's PM-WANI public WiFi scheme aims to make public internet popular by promoting public WiFi hotspots through private participation. There seems to exist lot of private and public interest for this scheme to advance and succeed, however very less PM-WANI public WiFi hotspots are observed on ground. The reasons are worth exploring.

Justification for public internet

Mobile internet seems to be satisfying the basic internet user requirements to a large extent; however specific user requirements may still not be met.

Choice of internet depends on the quality (speed and response) of internet promised by the internet provider in internet plans. The quality promised in the plans are usually loosely defined because networks are spatially deployed, capacity constrained and cannot meet strict quality requirements for all users simultaneously when loaded. Mobile networks are most prone to not deliver quality internet due to dynamically changing radio links and network loads caused by moving users. So, mobile internet users though seem satisfied for basic requirements may be challenged as follows, thus justifying need for public internet.

- Users in transit: When not satisfied with mobile internet in transit, they look for public internet with required and reliable quality usable on pay as you go basis.
- Users at home: When not satisfied with mobile internet quality at home, they opt for home internet provided using a dedicated wired (telephone, cable, fiber, etc.) or wireless (wireless local loop, satellite etc.) link. Home internet creates a personal hot spot at home for the user typically promising more reliable internet quality for the paid price. Home internet comes at a higher price than mobile internet and may not be affordable for all. Users who cannot afford a private home internet, look for public internet with required quality near homes usable on a pay as you go basis.

Reasons for low adoption of public internet

The public in general are not aware about public internet as an option for obtaining internet services in public places at reasonable prices. Such options were available in 1990's and 2000's in the form of internet browsing centres but have now been forgotten in the world of mobile internet. The internet providers are benefitted due to this low awareness as it allows them to continue promoting mobile and home internet services which are personal services and are less demanding on resources of the internet provider. Public internet is more demanding on resources as it requires providing enterprise grade internet and the connection is prone to high internet utilization due to resource sharing by many simultaneous users. The economic calculations of providing public internet thus keep internet providers away from promoting public internet. This decision of internet providers is sustainable because of less competition due to monopoly of few internet providers in delivering all of enterprise, mobile and home internet services.

The governments intervention by launching the PM-WANI scheme attempts to promote public internet by involving private "aggregators" providing hardware and software

infrastructure for user authentication, admission and accounting to the small shop owners deploying PM-WANI hot spots. Even in this arrangement, the aggregators and shop owners as private players are together unable to influence the internet provider to reduce price of enterprise internet. The Government many times tries to intervene to cap the enterprise internet pricing for PM-WANI, however this seems to have little effect on the internet providers due to the market power they enjoy.

Proposal to improve adoption of public internet

The way forward requires creating a competitive free market for public internet to exist in parallel to existing market for mobile and home internet.

To do this, government may want to reduce interventions and increase scheme visibility and branding by (i) Modifying the PM-WANI architecture to keep the internet provider in central role by absorbing the aggregator's role (ii) Letting PM-WANI user requests be directly handled by internet provider to understand public WiFi market demand (iii) Allowing internet providers to come up with innovative revenue sharing business models for the enterprise internet connections specific to PM-WANI (iv) Allowing shop owners to directly work with internet providers to adopt a suitable business model for setup of PM-WANI hotspots (v) Not putting any price caps on enterprise internet or retail internet and allowing the prices to settle down based on market demand and supply (vi) Make PM-WANI hotspots prominently visible with name boards like "PM-WANI Public WiFi" at PM-WANI hotspots across India (vii) Pilot the scheme and create competition in rural areas first where the need for public internet is expected to be more due to low mobile internet quality and low economic status of people making private home internet connections unaffordable.

Conclusion

Public internet does seem to have a market in India whenever mobile internet fails to deliver the required internet quality. However, limited user awareness, self-interest of monopolist internet providers, and more than needed government interventions have resulted in low adoption of public internet. Some minor changes in architecture of PM-WANI scheme and increased visibility of the scheme through appropriate branding may help to create the needed incentives to stake holders to make the scheme viable and successful.